

质数筛应用

人员

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上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4813>

王向东老师周日三点半C++并查集											
刷新											
#	用户名	姓名	编程分	时间	A	B	C	D	E	F	G
1	liuliwei	柳力玮	700	55084	100	100	100	100	100	100	100
2	mingzongliang	明宗亮	606	21220	100	100	100	100	100	100	6
3	liuqi	刘祺	600	4798	100	100	100	100	100	100	
4	zhaoshufan	赵书梵	500	3567	100	100	100	100	100		
5	yuanzhao	苑钊	500	6742	100	100	100	100	100		
6	yangyongcheng	杨咏丞	500	9502	100	100	100	100	100		
7	liruihan	李瑞涵	500	10963	100	100	100	100	100		
8	tianxinyi	田心一	500	11105	100	100	100	100	100		
9	gaojianhuan	高健桓	500	27148	100	100	100	100	100		
10	jibohan	纪博涵	490	8461	90	100	100	100	100		
11	jiangshuzhang	蒋叔璋	400	9298	100	100	100	100			
12	liuchenxi	刘宸熙	400	27103	100	100	100	100			

本周作业

<https://cppoj.kids123code.com/contest/4953> (课上讲了 A ~ E 题, 课后作业是 A ~ E 题必做, F G 选做)

课堂表现

今天讲了一些 质数筛 的做题应用, 同学们课上整体吸收还是不错的。

课堂内容

质因数分解

质因数分解模板

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

signed main()
{
```

```

int n, m; cin >> n;
m = n;
vector<int> vec;
for (int i = 2; i*i <= n; ++i) {
    while (n%i == 0) {
        vec.push_back(i);
        n /= i;
    }
}
if (n != 1) vec.push_back(n);

cout << m << "=";
for (int i = 0; i < (int)vec.size(); i++) {
    if (i) cout << "*";
    cout << vec[i];
}
cout << endl;
return 0;
}

```

乘积的约数个数

先做质因数分解, 然后可以求约数个数

```

#include <bits/stdc++.h>
#define int long long

using namespace std;

const int mod = 1e9 + 7;

signed main()
{
    map<int,int> mp;
    int n; cin >> n;
    while (n -- ) {
        int x; cin >> x;
        for (int i = 2; i*i <= x; ++i) {
            while (x%i == 0) mp[i]++, x /= i;
        }
        if (x != 1) mp[x]++;
    }

    int res = 1;
    for (auto it : mp) res = res*(it.second+1) % mod;
    cout << res << endl;
    return 0;
}

```

一定是 $\min(p,q) \leq 1e6$, 因此可以枚举 $\min(p,q)$ 的值

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 3e6 + 5;
bool st[maxn];
vector<int> primes;

void solve() {
    int x; cin >> x;
    for (int i : primes) {
        if (x%i == 0) {
            int t = sqrtl(x/i);
            if (t*t == x/i) {
                cout << t << " " << i << endl;
                break;
            }
        }

        if (x%(i*i) == 0) {
            cout << i << " " << x/(i*i) << endl;
            break;
        }
    }
}

signed main()
{
    for (int i = 2; i < maxn; ++i) {
        if (st[i]) continue;
        primes.push_back(i);
        for (int j = i+i; j < maxn; j+=i) st[j] = true;
    }

    int T; cin >> T;
    while (T -- ) solve();
    return 0;
}
```

Div Game

先做质因数分解, 然后根据分解后的指数来处理

```
#include <bits/stdc++.h>
#define int long long

using namespace std;
```

```

const int mod = 1e9 + 7;

signed main()
{
    map<int,int> mp;
    int n; cin >> n;
    for (int i = 2; i*i <= n; ++i) {
        while (n%i == 0) mp[i]++, n /= i;
    }
    if (n != 1) mp[n]++;

    int res = 0;
    for (auto it : mp) {
        int x = it.second, sum = 0;
        for (int j = 1; ; ++j) {
            sum += j;
            if (sum > x) {
                res += j-1;
                break;
            }
        }
    }
    cout << res << endl;
    return 0;
}

```

Coprime 2

对每个 a_i 做质因数分解, 找出来那些与 a_i 不互质的数, 剩下的就是互质的

```

#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 1e5 + 5;
bool st[maxn];

signed main()
{
    int n, m; cin >> n >> m;

    set<int> s;
    while (n -- ) {
        int x; cin >> x;
        for (int i = 2; i*i <= x; ++i) {
            while (x%i == 0) s.insert(i), x /= i;
        }
        if (x != 1) s.insert(x);
    }
}

```

```

for (int i = 1; i <= m; ++i) st[i] = true;
for (int i : s) {
    for (int j = i; j <= m; j += i) st[j] = false;
}

vector<int> vec;
for (int i = 1; i <= m; ++i) {
    if (st[i]) vec.push_back(i);
}

cout << vec.size() << endl;
for (int i : vec) cout << i << endl;
return 0;
}

```

250-like Number

先把 $\leq 1e6$ 的质数筛出来, 然后用 枚举+二分 优化

```

#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 1e6 + 5;
bool st[maxn];
vector<int> vec;

signed main()
{
    for (int i = 2; i < maxn; ++i) {
        if (st[i]) continue;
        vec.push_back(i);
        for (int j = i+i; j < maxn; j += i) st[j] = true;
    }

    int n, len = vec.size(), res = 0; cin >> n;
    for (int i = 0; i < len; ++i) {
        int a = vec[i];
        int l = i+1, r = len-1;
        while (l <= r) {
            int mid = (l + r) / 2;
            int b = vec[mid];
            if (a*b*b <= n/b) l = mid+1;
            else r = mid-1;
        }
        res += r-i;
    }
    cout << res << endl;
}

```

```
    return 0;  
}
```